Shreyas Popullu Jagadeesha Reddy

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EDUCATION

M.S. Computer Science 08/2023 - 05/2025
Arizona State University, Tempe, AZ
Relevant Coursework: Cloud Computing, Learning Methods in AI/Reinforcement Learning, Operational Deep Learning 4.0/4.0 GPA

B.S. Information Technology 07/2019 - 07/2023
International Institute of Information Technology, Bhubaneswar, India
Relevant Coursework: Data Structures and Algorithms, Databases, Operating Systems, Object Oriented Design 8.87/10 GPA

TECHNICAL SKILLS

Programming Languages and Front-End: Python, C, C++, Java, R, JavaScript, HTML, CSS, jQuery, React, Tkinter GUI
Frameworks, Databases and OS: TensorFlow, Keras, Pytorch, Scikit-Learn, Pandas, MySQL, MongoDB, PostgreSQL, Redis, Tableau, GenAI, FastAPI, Flask, Hadoop, OpenCV, Spark, Databricks, Git, Docker, Kubernetes, AWS, Linux

PROFESSIONAL EXPERIENCE

Centene Corporation, MO : Machine Learning Engineer II 05/2025 - Present
- Developed an API for NLP key-value extraction from unstructured, multimodal documents, reducing manual entry time by 70%
- Integrated the API with existing claims processing workflows, enabling automated extraction from 7M+ documents monthly
- Refactored a legacy OCR solution with memory optimizations reducing memory footprint by 45% while handling 3x larger batches
- Redesigned API's data persistence layer with caching and fallback mechanisms, reducing output latency by 40% with 99.5% uptime

Centene Corporation, MO : Machine Learning Engineer Intern 05/2024 - 05/2025
- Designed and deployed an API from scratch that classifies 16 medical document types based on images received via request
- Retrained a model classifier that achieved SOTA results with 94% accuracy using regional stacking/generalization of images
- Optimized this model to outperform previous model by 4% while reducing parameters by 63% and input image resolution by 25%
- Reduced the response time by 64.7% to meet SLA API standards

New York University, NY: Research Intern 06/2021 - 02/2022
- Evaluated the performance of 5 deep learning architectures namely RNN+LSTM, Dilated LSTM, CNNs and others using **Tensorflow** and **Keras** on NinaPro database for robotic hand prosthesis
- Spearheaded the creation of the **Dilated Efficient Capsule Network**, a novel deep learning architecture that outperformed the existing models by 10% and accomplished accuracy of 80% on 17 gestures with only 20% of original training signal data
- Published a paper titled "Transfer Learning of Dilated Efficient CapsNet: Towards Generalization for Neuro-robotics" in the IEEE RAL journal, accepted for presentation at IROS conference in Kyoto, Japan (October 2022) (link)

RELEVANT PROJECTS

img2speech python package, *Personal Project* *(link)*
- Launched an open-source **Python** package, enabling one-line conversion of textual images and PDF files to human audio files revolutionizing accessibility by providing audio content for visually impaired individuals
- Reduced the word error rates by an average of 5% by leveraging the built-in image preprocessing module and authored the package, now used by over 4235 developers and students in academic and research settings

Reliable AI Proctoring Software, *Personal Project* *(link)*
- Developed a remote proctoring software using **MediaPipe** and Pillow for video processing, PyAudio for audio processing and **NLTK** for text analysis to mitigate spoofing attempts and ensure secure online exams
- Orchestrated a project presentation that garnered over 3000 views on YouTube, showcasing the project's utility

Emotion based music recommendation system, *Personal Project* *(link)*
- Engineered a recommendation algorithm using **Python** that utilized user mood data to curate customized music playlists
- Extracted music features from 160,000 songs on the Spotify API data and performed data clustering to identify user preferences
- Achieved 105+ stars, 40 forks and 8500+ views on GitHub for the novel recommendation system using **Tkinter** GUI

Nina Funcs python package, *Research Project* *(link)*
- Devised a **Python** library for preprocessing and training of the EMG signals from the NinaPro database
- Implemented signal processing functions like filtering, windowing, rectification and PCA in 1 line of code and produced the package on PyPI and has over 4510 downloads

EXTRACURRICULAR EXPERIENCE

Udemy, Remote: Instructor, Data Structures and Algorithms in C++ (link) 01/2023 - 07/2023
Pioneered a highly-rated Udemy course on **Data Structures** and **Algorithms** in **C++**, boasting a 5-star rating for 14 consecutive months and enrollment of over 12500 students worldwide.

Kaggle, Data Science Company, Remote: Kaggle Master Tier (link) 05/2021 - 05/2022
Distinguished on Kaggle as a Master-tier contributor, ranking in the top 0.5%, demonstrated leadership by publishing innovative work having over 1300 forks and 70000 views on aggregate